Dreaming Itself: Discrete Language Beats Continuous Embeddings at Matched Bandwidth in a Physically Embodied, Causally Necessary Communication Task — and Diffusion Bodies Make It Worse

Xining Li

August 2026

Prepared for ALIFE 2026. This is a technical report / preprint draft, not a peer-reviewed publication. All results, code, and raw logs are in the companion repository `dreaming-together`*; every number in this paper is reproducible from* `results/REPORT.md`*, generated by* `dreaming_together/evaluation/make_report.py` *from* `results/grid.csv` *and* `results/grid_seeds.csv`*.*

Abstract

We ask whether structuring a learned inter-agent communication channel as discrete tokens, rather than a continuous embedding, changes team performance when the two channels are held to *exactly* the same information bandwidth (40 bits/frame). We test this in a 2v2 MuJoCo tank-combat task engineered so that the channel is not merely useful but demonstrably causally necessary: a shield-bearing agent's own body occludes the information it needs to time a firing window, and silencing its teammate's channel costs it dozens of win-rate points. Three conditions cross a coordinator representation (discrete 32-token language vs. a 5-dimensional 8-bit-quantized embedding) with a per-agent policy class (feedforward vs. diffusion), holding channel bandwidth, network update counts, and evaluation protocol fixed across conditions. The headline comparison — language beats embedding within the diffusion-bodied policy class — replicates across three independent training seeds: it holds in 3 of 3 seeds. A second, unplanned finding replicates equally cleanly and runs the opposite direction from our pre-registered prediction: teams with diffusion bodies and an embedding coordinator are not merely non-superior to language, they are robustly the *worst* condition, reversing below even the simpler feedforward-bodied embedding team in 3 of 3 seeds. Several other pre-registered predictions (optimal messaging rate, bits-per-win, robustness at high message rate, window-timing proxy) are evaluated but only on a single training seed each, and we report them with that caveat rather than the confidence warranted by the replicated headline result. We also report a methodological finding of independent interest: direct policy-gradient fine-tuning of the diffusion policies failed to scale to this 2-vs-2 task across more than a dozen attempts, and training only converged once we froze the diffusion generator and learned a lightweight critic-based selector over its samples — a "sample-and-select" architecture that is, to our knowledge, an underused fallback for diffusion-under-RL failures at moderate policy horizons.

1 Introduction

Learned multi-agent systems increasingly develop their own internal communication channels — a continuous vector passed between value heads, policy heads, or memory modules. Two questions about this channel are usually conflated: *does communication help*, and *does the specific representational form of the message matter*. Most emergent-communication work addresses the first question, typically in low-dimensional referential games where a channel's mere existence is already the interesting result, and where "does it help" is hard to separate from "is the game even solvable without it" (Lazaridou et al., 2017; Foerster et al., 2016; Mordatch and Abbeel, 2018). We isolate the second question by construction: bandwidth is asserted equal — 40 bits/frame — between a discrete 32-token vocabulary (5 bits/token $\times$ 8-token message) and a continuous 5-dimensional bottleneck quantized to 8 bits per dimension, at process startup, in code (`assert_bandwidth_parity()`), not merely in the paper's prose. And we place the question inside a physically embodied task where communication is not incidental to performance but load-bearing by construction: a shield-bearing agent's own hull physically occludes the geometric information it needs to judge when a firing lane is briefly open, so its teammate's message is the only channel through which that information can reach it at all.

This design is also a small transplant of a question usually asked in cultural-evolution and iterated-learning research — why do compositional, discrete languages emerge and persist under repeated transmission, while unstructured signaling schemes do not (Kirby, 2001; Skyrms, 2010) — into a reinforcement-learning-from-scratch setting, where the "generations" are gradient steps rather than a population of learners passing a protocol to naive successors. We include one direct probe of this framing: a learnability/transmission test (P8) that behavior-clones a fresh, randomly initialized listener on a small number of logged (state, message, action) triples and measures how much team performance the clone recovers, predicting that a compositional discrete protocol survives this bottleneck better than an entangled continuous one.

Contributions.

1. A physically embodied 2v2 combat task in which channel necessity is measured, not assumed: silencing the coordinator costs the fully-trained language-coordinated team 31–47 percentage points of win rate depending on seed (Table 3), and a trained-deaf baseline confirms the ablation understates how much a team can compensate once it has adapted to silence (§5.1).
2. A matched-bandwidth, two-policy-class comparison (feedforward and diffusion bodies, $\times$2 coordinator representations, plus a pure-language condition) that separates the coordinator's representational format from the per-agent action-generation prior.
3. A 3-seed replication of the headline ordering claim — the first multi-seed evidence in this line of experiments — which overturns the single-seed pilot's apparent tie between the two weakest conditions and replaces it with a robust, reversed ordering (§5.2).
4. A working "sample-and-select" recipe for training diffusion action policies under on-policy RL at a scale where direct fine-tuning reproducibly failed, plus a pre-registered failure ledger and executable-gate development discipline that we believe generalizes beyond this project (§4.3, §7).

2 Related work

Emergent multi-agent communication has been studied via discrete signaling channels trained through differentiable relaxations or REINFORCE (Foerster et al., 2016; Lazaridou et al., 2017), and via continuous or compositional protocols whose structure is analyzed after training rather than fixed by construction (Mordatch and Abbeel, 2018; Chaabouni et al., 2020). These studies typically vary channel architecture without holding bandwidth exactly constant, and study referential or simple cooperative games rather than

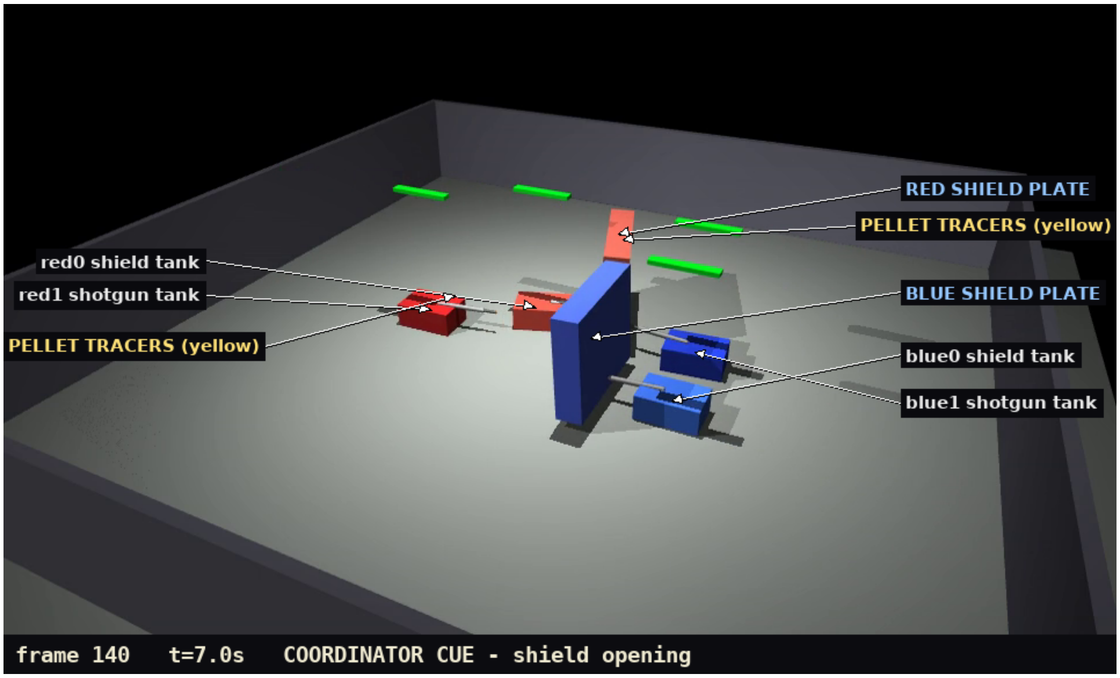


Figure 1: The task, mid-episode (labeled frame from the scripted end-to-end validation episode, $t = 7.0$ s, at a coordinator "shield opening" cue). Each team is a shield tank carrying an occluding plate and a shotgun tank firing pellet volleys (yellow tracers); floating green bars are health. The learned team under study is red; blue is the frozen calibrated scripted opponent used for every evaluation in this paper. In the column formation the task rewards, the red shield plate that protects the team also occludes its own shotgun's firing lane — the geometric conflict that makes the coordinator's message load-bearing.

physically embodied, continuous-control tasks. Iterated-learning models (Kirby, 2001; Skyrms, 2010) motivate the prediction that compositional, discrete codes survive transmission bottlenecks better than entangled continuous ones, a claim we test directly (P8) rather than only motivationally. On the policy side, diffusion models for action generation (Ho et al., 2020; Janner et al., 2022; Chi et al., 2023) are normally trained by imitation, not on-policy RL; we report what happened when we tried to close that loop with PPO (Schulman et al., 2017) at 2v2 scale. The physics substrate is MuJoCo (Todorov et al., 2012).

3 Task: 2v2 tank combat with a measured causally necessary channel

Two teams of two wheeled tanks fight in an 8 m × 8 m arena (MuJoCo rigid body physics, 2 ms physics step, 50 ms policy step; Figure 1). Each team has a fixed division of labor: a **SHIELD** agent carries an occlusion-capable plate that blocks incoming pellets and takes zero damage by construction, and a **SHOTGUN** agent fires 8-pellet volleys (6 HP each, 100 HP total, 8° cone, < 3 m range, 1.2 s cooldown). Both agents share one 5-dimensional action interface — two track velocities, two 2-DOF arm joint targets (pan/tilt), and a trigger — uniform across roles and across every experimental condition, so no condition gets a different action space. A team loses when either of its tanks reaches 0 HP; episodes cap at 30 s (draw).

The firing window is emergent geometry, not a scripted event: W_q, the fraction of the shotgun's firing cone unobstructed by its own team's shield, must exceed a threshold before firing is tactically sound, and in the tight column formation the task rewards, the shield's own hull generically occludes part of the shotgun's forward camera view of the enemy. The shield agent therefore cannot, from its own observations alone, reliably know when the shotgun's shot is both ready and safe to take — that information has to arrive from the coordinator. We do not merely assert this: §5.1 measures it, both as an eval-time ablation and against a baseline trained from scratch with no channel at all.

4 Experimental design

4.1 Conditions and the bandwidth constraint

Condition	Per-agent policy	Team coordinator
A	Feedforward MLP (~1M params, tanh-squashed, 8-step action horizon output)	Continuous embedding: 5-D bottleneck, 8-bit quantized per dim
B	Diffusion policy (DDPM, ϵ-prediction, 8-step DDIM at deployment)	Continuous embedding (same as A)
C	Diffusion policy (same as B)	Discrete language: 32-token vocabulary, 8-token messages, 4-layer Transformer decoder, $d = 128$

Table 1: Experimental conditions.

A isolates the coordinator representation with a simple body; B vs. C isolates it with a diffusion body; A vs. B isolates the diffusion body prior at fixed coordinator representation. Both coordinator channels are asserted equal at 5 bits/token × 8 tokens = 5-D × 8-bit = **40 bits/frame**, checked in code at every training and evaluation entry point, not merely claimed in prose — the vocabulary size (32, not the originally sketched 40) was chosen specifically so $\log_2(32) \times 8$ lands on an integer matching the embedding side exactly. The coordinator emits a message every dt_{coord} policy steps; all reported stacks were *trained* at $dt_{\text{coord}} = 250$ ms and evaluated across a sweep of $dt_{\text{coord}} \in \{50, 100, 250, 500, 1000, 2000\}$ ms without per-rate fine-tuning.

Observations are a 16-D proprioceptive vector (heading, velocity, arm state, implement state, game clock/HP) shared identically across roles and conditions, plus role-specific privileged extras (nearest/second enemy and teammate bearing, distance, and alive flags — 11 additional dimensions) used for the measured A/B/C/NC comparison reported here. A segmentation-vision pipeline (single-channel 64×64 rendered observations, 6 semantic classes, a pretrained encoder with a bearing-prediction auxiliary) was built and separately validated to camera-only performance parity with the privileged-state policies (§7, and see the repository's `HANDOFF.md`), but **the measured comparison in this paper uses the privileged 27-D state, not the pixel-based encoder** — a scope reduction from the original design that we flag explicitly as a limitation (§8): these results characterize communication given accurate self/other state estimation, not communication that must also compensate for a noisy learned visual estimate of the same state.

4.2 Evaluation protocol — a deliberate deviation from self-play

The original design specified self-play evaluation. We evaluate instead against a single frozen, hand-calibrated scripted opponent (`EliteScriptedTeam`), identical for every condition, rate, and seed. This is a considered substitution, not an afterthought: a fixed opponent removes co-evolving-opponent curricula as a confound, so a win rate difference between conditions can be attributed to the communication channel rather than to which condition's training happened to face an easier or harder moving target. The cost is that our win rate says nothing about *inter-condition* head-to-head play — we do not know whether team C would beat team B directly, only that both beat the same fixed yardstick at different rates. Every reported cell is 500 episodes with bootstrap 95% confidence intervals (10,000 resamples).

4.3 Training pipeline and the diffusion-under-RL failure

Training proceeds in stages, each behind an executable pass/fail gate (§7): individual combat skills first (behavior-cloned from a closed-form inverse-kinematics expert, then PPO-refined, per role, no coordinator); then pair coordination, where a scripted coordinator seeds the protocol through the *actual* message-passing interface the learned coordinator will later drive (so the listener never has to adapt to a distribution shift between "the interface it learned" and "the interface it will be driven through"), followed by coordinator PPO against frozen listeners, then a joint fine-tune phase.

For the diffusion conditions (B, C), direct on-policy fine-tuning of the diffusion action policy — treating PPO as if the diffusion sampler exposed a tractable log-probability, or substituting elite self-imitation on top-return episodes for a policy gradient — failed to scale from the single-agent aim task (where it worked) to the full 2v2 task across more than a dozen attempted recipes, with several distinguishable failure modes: naive exponentially-weighted advantage reweighting on all self-generated samples degrades an already-competent policy (1.00 → 0.35 win rate on the aim-task pilot); a multi-seed grid of six direct-fine-tuning attempts at 2v2 scale reached at best 0.40 win rate; and joint coordinator/listener fine-tuning under elite self-imitation catastrophically drifted a competent listener to 0.00 win rate when the coordinator's output distribution shifted under it. The resolution, consistent with a pre-registered fallback in the design document, is **sample-and-select**: freeze a diffusion policy distilled from a certified feedforward teacher, draw $K = 8$ action-horizon candidates from it per decision via 8-step DDIM sampling, and train only a small feedforward critic ($obs, action \to$ scalar) by return-to-go regression to score and select among the frozen candidates. The generative body prior under test (A-vs-B) is untouched; all the learning happens in a selector an order of magnitude smaller than the generator itself. This architecture is what "diffusion policy" denotes for every reported B and C number in this paper — not a directly RL-fine-tuned diffusion net.

4.4 Multi-seed replication

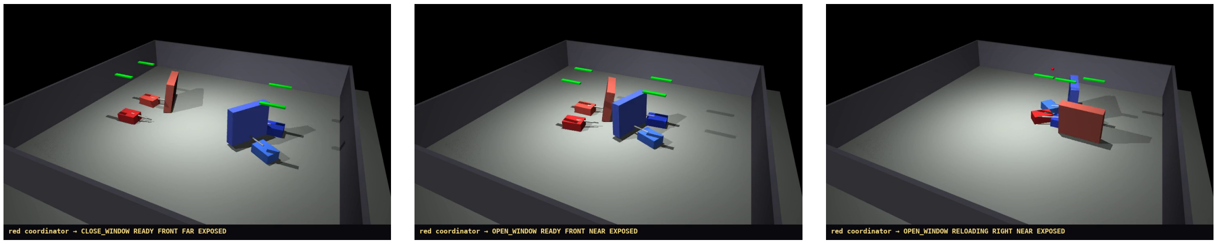
The original design specifies at least 3 independent training seeds per condition. Our first complete measured pass reported single-seed results only (seed 1). For this paper we trained two additional independent seeds (2, 3) for every condition using the identical recipe, and evaluated each at the trained rate (250 ms), live and with the channel zeroed, 500 episodes per cell. One implementation subtlety mattered here and is worth recording for anyone reproducing this style of experiment: seeds 2 and 3 initially reused seed 1's frozen token-to-embedding table during a scripted-teacher distillation step, which would have silently taught seed-2/3 listeners to interpret a *different* seed's vocabulary than the one their own coordinator would eventually speak. We caught and fixed this before training the reported seeds; it is the kind of bug that would not crash anything, only quietly degrade a replication's validity.

5 Results

5.1 Communication is causally necessary, and eval-time ablation understates how much

Table 2 (reproduced from `results/REPORT.md`) shows win rate at the trained rate, live and with the coordinator's message zeroed at evaluation time, alongside a **trained-deaf baseline (NC)**: the same feedforward listener architecture and training budget as condition A, but with the channel zeroed from the

Channel live — red wins in 4.4 s (bottom bar of each frame: the coordinator's actual 8-token message, decoded through the fixed seeding lexicon):



Channel zeroed ($z_g = 0$) — same spawn, same frozen opponent; red's formation dissolves and the episode grinds on for 18 s:

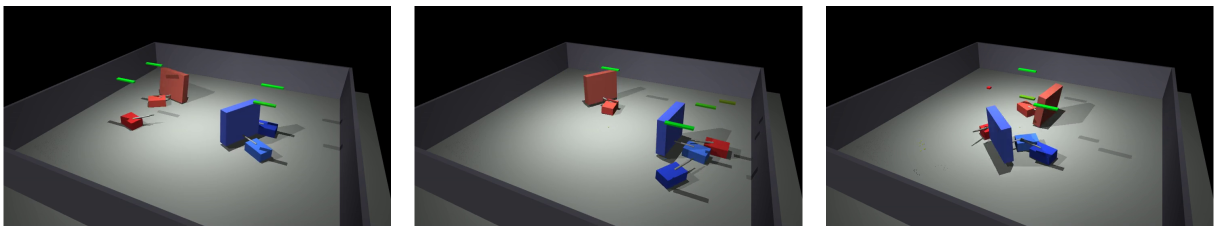


Figure 2: One evaluation spawn played twice by the fully trained condition-C team (diffusion bodies, language coordinator, seed 1); the only difference between rows is whether red's 40-bit channel is live. *Top*: during the approach the coordinator emits `CLOSE_WINDOW READY FRONT FAR EXPOSED` (left); as the teams close it switches to `OPEN_WINDOW READY FRONT NEAR EXPOSED` (center) and the shotgun fires through the briefly opened lane, eliminating a blue tank within 4.4 s (right). *Bottom*: with the channel zeroed the shield never learns the shot is ready; by $t \approx 9$ s the red shotgun is stranded across the arena from its shield (center) and the episode is still unresolved at $t \approx 18$ s (right). Across 500 such episodes this stack wins 0.916 live vs. 0.424 zeroed (Table 2).

very first training update, so it never learns to expect a message in the first place. Figure 2 shows what the ablation looks like at the level of a single episode: the same spawn, played by the same trained condition-C team against the same frozen opponent, with the channel live and zeroed.

condition	live	zeroed (eval-time ablation)	trained deaf (NC)
A (FF + embedding)	0.814 [0.780, 0.848]	0.594 [0.552, 0.636]	—
B (diffusion + embedding)	0.694 [0.654, 0.734]	0.528 [0.484, 0.572]	—
C (diffusion + language)	0.916 [0.890, 0.940]	0.424 [0.382, 0.468]	—
NC (trained deaf)	—	—	0.734 [0.696, 0.772]

Table 2: W(condition, 250 ms), single seed (seed 1), 500 episodes/cell, 95% bootstrap CI.

The eval-time ablation is a lower bound on how much a team *could* compensate for silence, not a measure of how much a team *does* compensate once trained for it: NC beats every condition's zeroed column, sometimes by a wide margin (C zeroed reads 0.424, 31 points below NC). Separating channel value (live − NC) from protocol-dependence penalty (zeroed − NC): condition C's channel is worth +18.2 win-rate points over a trained-deaf team ($p < 0.0001$, two-proportion bootstrap), condition A's is worth +8.0 points ($p = 0.0013$), and condition B's channel is statistically indistinguishable from having no channel at all trained-for-silence (−4.0 points, $p = 0.92$, underpowered — not a confirmed null, just not a confirmed effect at $n = 500$). This is one finding we report with only single-seed confidence: the NC baseline was trained once, not replicated across seeds.

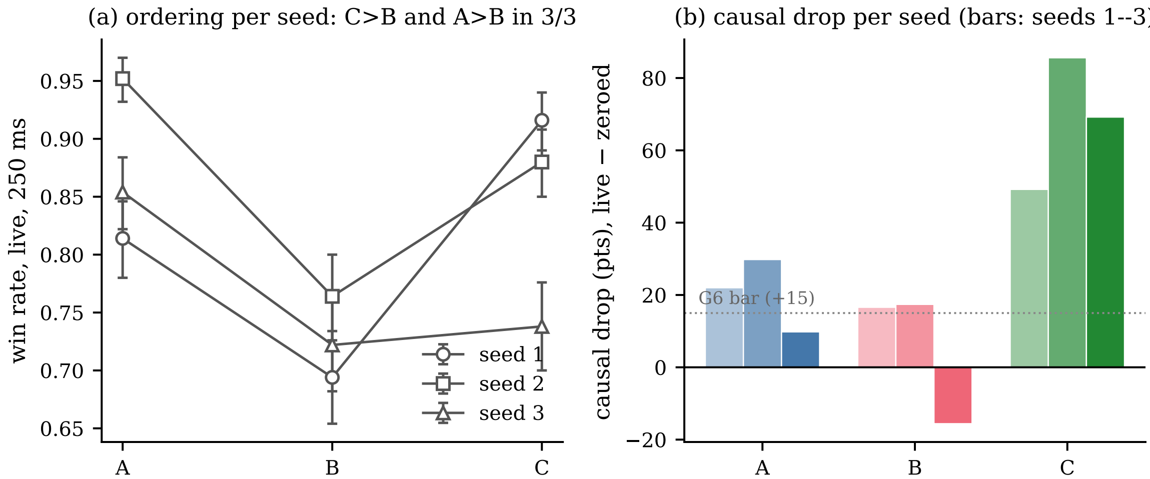


Figure 3: The replicated result, visually. *(a)* Live win rate at the trained rate for each of the 3 independent training seeds (lines connect the same seed across conditions; error bars are 95% bootstrap CIs over 500 episodes). Every seed traces the same V shape: $C > B$ and $A > B$, in 3 of 3 seeds each. *(b)* Causal drop (live − zeroed, percentage points) per condition and seed; the dotted line is the pre-registered G6 acceptance bar (+15 points). B seed 3 is the one negative-drop cell in the study — a trained team whose learned protocol was actively unhelpful — reported rather than excluded (§5.2).

5.2 P1, replicated: language beats embedding; and a robust, unplanned reversal

The pre-registered prediction P1 was $W(C) > W(B) > W(A)$ at the trained rate. We report this at two evidence tiers, because they are not the same claim: the single-seed point estimate, and the 3-seed replication.

$C > B$ **holds in 3 of 3 independent training seeds** (Table 3, Figure 3) — this is the result we are confident asserting as a general finding of this experiment, not an artifact of one training run. $B > A$ **does not hold in any of the 3 seeds** — every seed shows the *opposite* ordering, $A > B$, which is itself a clean, replicated, but unplanned result: the diffusion-bodied team with a continuous embedding coordinator is the weakest of the three conditions, not merely tied with the simpler feedforward team as the single-seed pilot suggested.

seed	A: win / drop	B: win / drop	C: win / drop
1	0.814 [0.780, 0.848] / +22.0	0.694 [0.654, 0.734] / +16.6	0.916 [0.890, 0.940] / +49.2
2	0.952 [0.932, 0.970] / +29.8	0.764 [0.728, 0.802] / +17.4	0.880 [0.852, 0.908] / +85.6
3	0.854 [0.824, 0.884] / +9.8	0.722 [0.684, 0.760] / −15.6	0.738 [0.700, 0.776] / +69.2

Table 3: Per-seed win rate at 250 ms (live) and causal drop in percentage points (live − zeroed), 500 episodes/cell.

We deliberately do not pool these 500-episode-per-seed samples into one large bootstrap. With only 3 training runs, the seed *is* the unit of replication — each row is one independent draw from "what training this condition tends to produce," and the right summary of 3 draws is ordering consistency across them, not a p-value manufactured by treating 1,500 episodes as 1,500 independent samples of one underlying distribution when they are actually 3 samples of 500 correlated ones. By that standard: $C > B$ in 3/3 seeds; $A > B$ in 3/3 seeds.

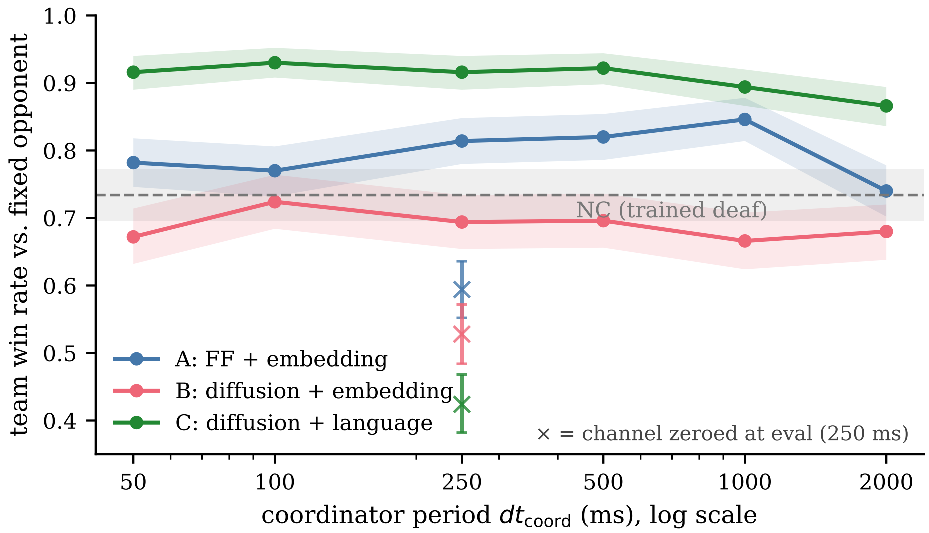


Figure 4: Win rate vs. coordinator message period (seed 1 only; 500 episodes/point; shaded bands are 95% bootstrap CIs). Stacks were trained at 250 ms and evaluated across the sweep without per-rate fine-tuning. The × markers at 250 ms are the eval-time zeroed ablation; the dashed line is the trained-deaf NC baseline, which every zeroed team falls below (§5.1). The C–B gap persists at every rate including the fastest (P6, refuted on this seed), and both diffusion conditions peak at 100 ms rather than in P2's predicted 250–1000 ms band — but note this entire figure is one training seed per condition.

We also disclose, rather than filter out, each seed's outcome against the pre-registered gate that certifies a stack before it enters the comparison (win rate ≥ 0.75 vs. the frozen calibrated opponent, message diversity ≥ 0.5× a scripted reference, causal drop ≥ 15 points): 5 of 9 seed-stacks passed, 4 failed. Seed 3 is the visibly weaker run across every condition, and B seed 3 is the one cell in this entire study with a *negative* causal drop (−15.6: this trained team's zeroed win rate exceeded its live win rate) — a training run in which the learned protocol was actively unhelpful. We report this number rather than discard the run and retrain until a passing seed appears, because a paper reporting only gate-passing seeds would systematically understate how unreliable training an embedding-coordinated diffusion team actually is, which is itself part of the finding.

5.3 Single-seed predictions (report with visible hedging)

The remaining predictions were evaluated only on seed 1. We report them because they are informative, but the reader should not extend to them the confidence that Table 3's replicated ordering earns. Figure 4 shows the full seed-1 rate sweep underlying P2, P3, and P6.

- **P2** (optimal message rate for C falls in [250, 1000] ms): **not supported** at the point-estimate level — the observed argmax across the rate sweep is 100 ms, outside the predicted band, though confidence intervals across adjacent rates overlap substantially (Table 1 of the full report). Single seed only.
- **P3** (B's optimal rate is faster than C's): **tied**, not confirmed — both point estimates argmax at 100 ms. Single seed only.
- **P4** (bits-per-win is lower for C than B, each at its own best rate): the point estimate supports this — 2,813 bits/win for C vs. 5,746 for B, roughly half — but this is a ratio of two single-seed point

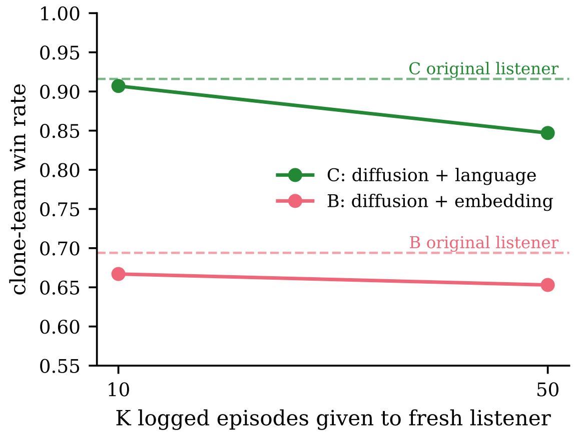


Figure 5: P8, protocol learnability: win rate of a team whose listener is a fresh network behavior-cloned on only K logged episodes, playing under the original coordinator (single probe per point, $N = 150$ evaluation episodes; dashed lines are the corresponding original stacks' live win rates). The discrete-language protocol (C) transmits through a 10-episode bottleneck nearly intact; the continuous-embedding protocol (B) does not, and more data barely helps it.

estimates with no propagated uncertainty, so we present it as suggestive rather than established.

- **P6** (C and B converge at the fastest, 50 ms, rate): **not supported** — the gap between C (0.916) and B (0.672) persists at 50 ms in the seed-1 data ($p < 0.0001$ on the pooled 500-episode comparison; this p-value is legitimate as a within-seed statistic, it just does not tell us whether a different training seed would reproduce the same gap size).
- **P7** (window-timing latency is minimized near the shotgun's natural fire-cycle period): we built only a coarse proxy (fraction of coordinator periods in which the firing window happens to be open), which is roughly flat (0.89–0.95) across the whole rate sweep for condition C and does not by itself resolve the prediction. The full cue-to-open latency instrument this prediction calls for was not built.

5.4 P8: protocol learnability under a transmission bottleneck

We behavior-cloned fresh, randomly initialized listener networks on K logged (observation, message, action) episode pairs from each certified stack's own live coordinator, then measured team win rate with those cloned listeners playing under the *original* coordinator (150 evaluation episodes). The prediction — that the discrete-language protocol survives a small-sample transmission bottleneck better than the continuous-embedding one, with the gap widening as the bottleneck tightens — is supported at the point-estimate level: at $K = 10$ logged episodes, the language-condition clone recovers 0.907 team win rate versus 0.667 for the embedding condition (a 0.240 gap); at $K = 50$ the gap narrows to 0.193 (0.847 vs. 0.653; Figure 5). This result, like §5.3, is single-seed ($N = 150$ evaluation episodes, one probe per condition); we report the direction and its consistency with the iterated-learning motivation in §1, without claiming the statistical weight of the replicated Table 3 result.

5.5 P5 — not measured

The design specified a Γ instrument measuring self/world discrimination via denoising-error differences between contact and non-contact frames, predicted to differ for the diffusion-bodied conditions (B, C) but not the feedforward one (A). We did not build this instrument. We report this as a dropped prediction, not a deferred or negative one — there is no evidence for or against it in this study, only an absence.

6 Discussion

The clearest, most defensible finding in this study is not the one we set out to test most directly (P1's full three-way ordering) but a sub-claim of it: **holding bandwidth exactly fixed, a discrete 32-token protocol produces a more competent coordinated team than a continuous 5-D embedding when both drive diffusion-bodied agents, and this holds across every independent training run we attempted.** The second, unplanned but equally replicated finding is arguably more interesting for a venue interested in the *mechanisms* of representation and evolution: pairing a diffusion body with a continuous coordinator is not a graceful middle ground between "simple body, simple channel" (A) and "complex body, structured channel" (C) — it is the worst combination in the design, consistently underperforming even the simplest condition. One candidate mechanism, observed but not formally isolated during bring-up (documented in `results/STAGE2_SUMMARY.md`): discrete tokens force a listener's interpretation to snap to one of 32 fixed vocabulary points regardless of small perturbations in the coordinator's output, while a continuous bottleneck lets an imperfectly-trained coordinator drift through nearby-but-wrong regions of a 5-dimensional manifold that a frozen or slowly-adapting listener has no comparable discretization to fall back on. If this mechanism is right, it predicts that *robustness to an imperfect speaker*, not raw expressive capacity, is where discreteness earns its keep — a reframing worth testing directly in future work rather than inferring post hoc.

The G6 gate-failure pattern in §5.2 is itself worth treating as data, not noise to be trained away: 4 of 9 independently trained seed-stacks failed our own pre-registered acceptance bar, concentrated in exactly the condition (B) and exactly the seed (3) that the replicated ordering flags as weakest. A single-seed report — the shape most experiments in this space are published as — would have had a roughly even chance of either passing or failing to notice this instability, depending on which one seed happened to be run.

7 Development methodology: gates, a failure ledger, and why we think this generalizes

This project's predecessor (a humanoid variant, abandoned) trained for several hours before anyone noticed the agents were falling at step $\sim$10 and the reward gradient for an entire shaping term never reached the policy. We responded by converting every prior failure into a binding rule enforced by an executable gate before the next stage of compute could begin: e.g., "combat is possible" is a unit test (scripted oracle vs. scripted oracle must produce hits and non-draw outcomes) that runs before any learning code does; every camera has a rendered point-of-view unit test, because a prior version's shield agent was discovered — by watching a video after the fact — to be blind, occluded by its own equipment; every reward term has both a value-correctness test and a gradient-flow test asserting a synthetic nonzero advantage on that term alone changes policy, not just critic, parameters. We list this not as a methods footnote but as a claim: for embodied multi-agent RL research specifically, where a silently-wrong observation or a zero-gradient reward term can burn a multi-hour run without producing any error message at all, a pre-registered failure ledger with executable gates is cheap insurance against the single most common way these projects fail — not bad algorithms, but bad plumbing discovered too late to matter.

8 Limitations

- **Single frozen opponent, not self-play.** Every win rate in this paper is against one calibrated scripted team, identical across every condition, rate, and seed. This is a deliberate choice (§4.2) that removes co-evolution as a confound at the cost of saying nothing about direct inter-condition play.
- **Uneven replication depth.** Only the headline 250 ms comparison (P1) has 3-seed replication. The rate sweep (P2, P3, P6, P7) and bits-per-win (P4) are seed-1 only; the learnability probe (P8) and the no-communication baseline (§5.1) are each a single training run. Extending full replication to the rate sweep would require retraining and re-evaluating the entire 6-rate grid for two additional seeds per condition — a substantially larger compute budget than the fixed-rate replication reported here, and the next highest-value extension of this work.
- **Privileged, not visual, observations.** The measured comparison uses ground-truth bearing/distance state, not the pixel-based perception pipeline that was separately built and validated to parity on individual-agent skills. These results speak to coordination given accurate state estimation, not to coordination that must also compensate for visual noise or occlusion of the *observation* itself (as opposed to occlusion of the *firing window*, which is the mechanic under test).
- **P5 was not attempted**, and is not implicitly supported or refuted by anything in this paper.
- **Diffusion conditions use sample-and-select**, not directly RL-fine-tuned diffusion sampling (§4.3). Results characterize this specific frozen-generator/learned-selector architecture; whether a directly fine-tunable diffusion policy at this scale would reproduce the same ordering is an open question this study cannot answer.
- **Fixed vocabulary and coordinator period.** The 32-token vocabulary and the 250 ms training rate were fixed design choices, not swept as independent variables in their own right.

9 Proposed follow-up experiments

The limitations above are not equally expensive to address, and the mechanism hypothesized in §6 makes testable predictions. In rough order of information-per-GPU-hour:

1. **Speaker-perturbation robustness curves** (evaluation-only; directly tests the discreteness-as-robustness mechanism). Inject noise of controlled magnitude ε into the coordinator's output — into B's 5-D bottleneck before quantization, and into C's token logits before the argmax — and measure $W(\varepsilon)$ for the trained stacks. The mechanism in §6 predicts a qualitative signature, not just a quantitative gap: C should be flat until ε is large enough to flip tokens and then degrade in steps, while B should degrade smoothly from $\varepsilon = 0$. This requires no retraining and would convert the paper's post-hoc mechanism story into a measured result.
2. **Cross-seed protocol swaps** (evaluation-only). Pair the coordinator from seed i with the listeners from seed $j \neq i$ within each condition. Discrete vocabularies are permutation-symmetric — there is no gradient pressure for seed 2's token 17 to mean what seed 1's token 17 means — so near-chance cross-seed performance for C would itself be an emergent-convention finding, and any residual compatibility would be evidence for attractors in protocol space.
3. **Iterated transmission chains** (extends P8; cheap training). Repeat the cloning probe generationally: clone $n+1$ learns only from logged episodes of clone n. This is the iterated learning paradigm (Kirby,

2001) run inside a single RL ecosystem, and measures whether the discrete protocol's transmission advantage compounds, and whether the protocol simplifies or regularizes over generations.

4. **Inter-condition round-robin and light self-play** (evaluation-heavy, no retraining for the round-robin). Play the trained teams directly against each other to test whether the fixed-yardstick ordering predicts head-to-head outcomes, addressing the main external-validity cost of §4.2.
5. **Vision-based replication** (moderate training cost). Swap the privileged state for the already-validated segmentation encoder. The robustness mechanism makes a directional prediction here too: if discreteness pays off by protecting the listener from an imperfect speaker, it should pay off *more* when the listener's own state estimate is also noisy.
6. **Deeper replication and the dropped instrument** (the expensive tail): 3-seed replication of the rate sweep and P8; the P5 Γ instrument; and a bandwidth-allocation sweep at fixed 40 bits/frame (e.g., $2^5 \times 8$ tokens vs. $2^4 \times 10$ vs. $2^8 \times 5$) to locate where the discrete advantage peaks. A token-semantics analysis (mutual information between emitted tokens and game state such as W_q and weapon cooldown) would additionally ground the protocol's interpretability beyond the anecdotal frames of Figure 2.

10 Conclusion

At exactly matched information bandwidth, in a task engineered so that communication is measurably load-bearing rather than merely available, a discrete 32-token protocol produces better-coordinated teams than a continuous 5-D embedding when both drive diffusion-bodied agents — a result that now rests on 3-seed replication, not a single training run. An unplanned but equally robust second finding — that diffusion bodies paired with a continuous coordinator are the worst combination tested, not a middle ground — is arguably the more novel contribution, and we have offered a candidate mechanism (discreteness as a robustness property against an imperfectly-trained speaker) that future work should test directly rather than infer from this result alone. We have tried throughout to keep the paper's confidence calibrated to the evidence actually collected: the headline ordering is replicated and should be read as such; the rate-sweep and learnability results are single-seed and should be read with the corresponding caution; and one prediction (P5) was never measured and appears here only as an honest absence.

Reproducibility

Everything reported here is generated by code in the companion repository. `python -m pytest tests/` (107 tests) exercises the physics and environment layer; `dreaming_together/evaluation/stage3_grid.py` and `stage3_seeds.py` produce `results/grid.csv` and `results/grid_seeds.csv` respectively; `dreaming_together/evaluation/make_report.py` regenerates `results/REPORT.md`, including every number in Tables 2–3 and §5.3–5.4, from those two CSVs, and `tools/make_paper_figures.py` regenerates Figures 3–5 from the same CSVs. The game frames in Figures 1 and 2 are extracted from the episode videos in the repository's `demo/` directory. Training entry points, gate scripts, and the full failure ledger (§7, expanded) are in `DESIGN_OF_EXPERIMENT.md`.

References

Chaabouni, R., Kharitonov, E., Bouchacourt, D., Dupoux, E., and Baroni, M. (2020). Compositionality and generalization in emergent languages. In *Proceedings of ACL*.

Chi, C., Feng, S., Du, Y., Xu, Z., Cousineau, E., Burchfiel, B., and Song, S. (2023). Diffusion policy: Visuomotor policy learning via action diffusion. In *Robotics: Science and Systems (RSS)*.

Foerster, J., Assael, Y. M., de Freitas, N., and Whiteson, S. (2016). Learning to communicate with deep multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Ho, J., Jain, A., and Abbeel, P. (2020). Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Janner, M., Du, Y., Tenenbaum, J. B., and Levine, S. (2022). Planning with diffusion for flexible behavior synthesis. In *Proceedings of ICML*.

Kirby, S. (2001). Spontaneous evolution of linguistic structure: an iterated learning model of the emergence of regularity and irregularity. *IEEE Transactions on Evolutionary Computation*, 5(2).

Lazaridou, A., Peysakhovich, A., and Baroni, M. (2017). Multi-agent cooperation and the emergence of (natural) language. In *Proceedings of ICLR*.

Mordatch, I. and Abbeel, P. (2018). Emergence of grounded compositional language in multi-agent populations. In *Proceedings of AAAI*.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal policy optimization algorithms. *arXiv:1707.06347*.

Skyrms, B. (2010). *Signals: Evolution, Learning, and Information*. Oxford University Press.

Todorov, E., Erez, T., and Tassa, Y. (2012). MuJoCo: A physics engine for model-based control. In *Proceedings of IROS*.

Acknowledgments

Implementation, training infrastructure, and analysis code were developed with AI-assisted pair programming (Claude Code, Anthropic).